

Multimodal Well-Being Assessment App

Literature Survey

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Mental health problems such as stress, burnout and depression are rising around the world. A key difficulty is that many people do not notice their own emotional decline until it has become serious, and common assessments depend on infrequent self-reports that can be biased. This project is a mobile app that acts as a mental health monitoring tool. It uses AI to analyse three data streams (facial expressions, voice patterns and structured psychological questionnaires) to assess a user's state of mind. It tracks changes over time to spot early warning signs of burnout, depression, suicidal ideation and other conditions. A user records a short video or audio clip, answers a few questions, and receives a wellness score with a summary of possible risks.

Table of Contents

1.0 Introduction	4
2.0 Mental Health Assessment: Background and Limitations	4
2.1 Overview of Mental Health Assessment	4
2.2 Traditional Assessment Methods	4
2.3 Limitations of Self-Report and Infrequent Assessment	4
3.0 Facial Expression Analysis for Emotion and Mental State Detection	4
3.1 Fundamentals of Facial Expression Recognition and Affective Computing	4
3.2 Facial Action Coding and Emotional Taxonomies	4
3.3 Machine Learning and Deep Learning Approaches to FER	4
3.4 Evidence Linking Facial Cues to Depression, Stress and Burnout	5
3.5 Relevant Datasets	5
3.6 Limitations and Challenges	5
4.0 Voice and Speech Analysis for Mental State Detection	5
4.1 Acoustic and Prosodic Features Relevant to Mental Health	5
4.2 Speech Emotion Recognition Techniques	5
4.3 Evidence Linking Vocal Markers to Depression and Suicidal Ideation	5
4.4 Relevant Datasets and Corpora	6
4.5 Challenges	6
5.0 Psychological Questionnaires and Digital Assessment Tools	6
5.1 Standardized Instruments	6
5.2 Digitalization of Questionnaires in Mobile Health Apps	6
5.3 Validity and Reliability of Digital Self-Report	6
6.0 Multimodal Data Fusion for Mental Health Prediction	6
6.1 Rationale for Combining Multiple Modalities	6
6.2 Fusion Strategies	7
6.3 Existing Multimodal Mental Health Systems	7
6.4 Machine Learning Architectures Used for Fusion	7
7.0 Mobile Health (mHealth) Applications for Mental Wellness	7
7.1 Overview of Existing Mental Health Monitoring Apps	7
7.2 Comparative Strengths and Gaps	8
7.3 Design Considerations for AI-Based Mobile Health Applications	8
8.0 Ethical, Privacy and Safety Considerations	8
8.1 Data Privacy and Security	8
8.2 Ethical Concerns in AI-Based Screening	8
8.3 Responsible Handling of Suicidal Ideation Risk	8

9.0 Research Gap and Summary	8
9.1 Synthesis of Findings	9
9.2 Identified Gaps	9
10.0 References	10

1.0 Introduction

Stress, burnout and depression have been rising over the years (Biomedical and Pharmacology Journal, 2025b). The problem is that people often don't notice they are getting worse. Someone who is slowly burning out can still turn up to work or school every day, and nobody sees anything wrong. By the time they ask for help, or a friend or family member notices, things may already be serious. That is why we need screening tools that are cheap, easy to reach and quick to use (Frontiers in Psychiatry, 2022). A smartphone app is a good fit, since people can use it privately every day.

2.0 Mental Health Assessment: Background and Limitations

2.1 Overview of Mental Health Assessment

Stress, burnout and depression are linked, but each has its own signs. These show up in things like facial expression, sleep, speech and social activity (Biomedical and Pharmacology Journal, 2025b). If chronic stress and burnout go unnoticed, they can turn into clinical depression, and in the worst cases into suicidal thoughts. Catching problems early is therefore an important place to step in (PMC, 2026).

2.2 Traditional Assessment Methods

Professionals usually assess mental health through structured or semi-structured interviews. These are backed up by questionnaires such as the Patient Health Questionnaire-9 (PHQ-9) and the Generalized Anxiety Disorder-7 (GAD-7) (Kroenke, Spitzer and Williams, 2001). Both are still the most common screening tools in clinics and in apps, because they are proven to work and easy to give.

2.3 Limitations of Self-Report and Infrequent Assessment

The PHQ-9 and GAD-7 both depend on people being willing and able to describe their own symptoms honestly (Neuropsychiatric Disease and Treatment, 2021). Stigma, low self-awareness and the fact that clinic visits are rare all get in the way. Research on speech and suicide risk makes the same point: interviews and questionnaires only work if the person is ready to admit they are struggling. This is why researchers are looking for more objective signals that can be collected passively, without relying only on what people say about themselves.

3.0 Facial Expression Analysis for Emotion and Mental State Detection

3.1 Fundamentals of Facial Expression Recognition and Affective Computing

The idea behind facial analysis is simple. Facial muscle movements can be seen and measured, and they give clues about how someone feels inside. People with depression tend to be less expressive. They smile less, show more sadness and have a narrower range of emotions. Computer vision systems can learn to pick up these patterns straight from video (ScienceDirect, 2025a).

3.2 Facial Action Coding and Emotional Taxonomies

Most automated facial analysis rests on two things. The first is coding systems that split facial movement into small units called action units. The second is a set of basic emotion categories: happiness, sadness, anger, fear, surprise and disgust. Together they give deep learning models the labels they need to learn from images and video.

3.3 Machine Learning and Deep Learning Approaches to FER

A systematic review of deep learning for depression recognition (2017 to 2024) showed clear progress. Models that look at space and time across many video frames did noticeably better than models that use a single still image (ScienceDirect, 2025a). A common setup pairs CNNs with recurrent networks like LSTMs, so the model learns both what a face looks like and how it changes. In one study, LSTM networks with attention over action units, head pose and eye gaze were used to estimate depression severity from interview videos (MDPI/PMC, 2023). General emotion recognition systems also score very high on benchmarks like CK+, RAF-DB, FER2013 and FERPlus, so the technology is quite mature.

3.4 Evidence Linking Facial Cues to Depression, Stress and Burnout

There is direct evidence that facial movement is linked to depression. One study compared frame by frame a group simulating depression with a normal group. The simulated-depression group showed much less change in emotion over the recording, more sadness, and fewer neutral or fearful expressions (PMC, 2024a). This fits the known link between flat affect and depression. It also suggests that how expressive a person is can be a useful feature in itself, not just which emotion label they get.

3.5 Relevant Datasets

Two kinds of datasets are used:

- **General emotion datasets** (CK+, FER2013, FERPlus, RAF-DB), used for pretraining and benchmarking.
- **Clinical interview datasets** (DAIC-WOZ, E-DAIC), which pair interview video with validated depression scores. These are now the usual choice for training and testing models built for depression.

3.6 Limitations and Challenges

Face detection struggles with poor lighting, odd camera angles and anything blocking the face. People also show emotion differently across cultures and as individuals. The biggest problem is a lack of large, varied, clinically labelled video datasets. Recording and storing identifiable video of patients raises privacy issues, so models often don't generalise well in real use.

4.0 Voice and Speech Analysis for Mental State Detection

4.1 Acoustic and Prosodic Features Relevant to Mental Health

Researchers look at particular voice features, such as pitch, fundamental frequency, jitter, shimmer, speech rate, pauses, and glottal and phonation traits. These change when someone is anxious or depressed, partly because body movement and vocal control slow down (Journal of Technology in Behavioral Science, 2024).

4.2 Speech Emotion Recognition Techniques

Machine learning on acoustic and prosodic features gives good clues about mood changes and depression level. Studies suggest these signals can track how symptoms change over time. The catch is that each study uses a different setup, which makes it hard to move these tools into clinical practice. Deep learning methods such as CNNs, GRUs and BiLSTMs are common. Newer work uses pretrained models like Wav2Vec 2.0 to pull emotional information directly from audio (arXiv, 2025a).

4.3 Evidence Linking Vocal Markers to Depression and Suicidal Ideation

Recent work has moved from general depression toward suicide risk. One study used 588 narrative audio recordings sent through a mobile app by military veterans being screened for suicidal ideation. Using prosodic, phonation and glottal features, the researchers built a classifier and showed that smartphone audio works well for this task (BioData Mining, 2021). Another project applied natural language processing to interview text to separate high-risk from non-risk cases (PMC, 2023). Both found that what people say and how they say it carry useful signals. Other studies followed adolescents in treatment to track recovery, worsening or stable depression over time, and one used RNN classifiers on voice journals to tell manic from depressive moods (PMC, 2024b).

4.4 Relevant Datasets and Corpora

DAIC-WOZ is the most used corpus in voice studies, because it links audio with PHQ-8 severity scores. Other researchers build their own datasets, for example veteran phone recordings or clinical interviews. They do this mainly because large, public, clinically labelled speech datasets are rare.

4.5 Challenges

Background noise is a big problem for everyday mobile use. Different phone microphones, accents and languages make it hard for one model to work for everyone. Most studies use small samples and different testing methods, so results that look great in the lab often drop when tested on new people or new recording conditions.

5.0 Psychological Questionnaires and Digital Assessment Tools

5.1 Standardized Instruments

A meta-analysis found that the PHQ-9 works well on mobile phones and detects depression about as well as the paper version (Neuropsychiatric Disease and Treatment, 2021). It is widely used because it is so well validated. It is often paired with the GAD-7 to screen for anxiety, in clinics and in digital tools.

5.2 Digitalization of Questionnaires in Mobile Health Apps

Smartphone and desktop apps work well for tracking symptoms over time, and this fits with measurement-based care. Many mHealth platforms include the PHQ-9 to spot raised depressive symptoms in different countries and groups (Frontiers in Psychiatry, 2022). Many also use it together with the GAD-7 to measure depression and anxiety during remote check-ins.

5.3 Validity and Reliability of Digital Self-Report

Digital questionnaires have the same weaknesses as paper ones. Users may misremember, or answer in a way that looks good, and many stop answering after a while. On the other hand, digital forms offer more privacy and convenience. Researchers increasingly combine questionnaire answers with passive signals like facial expression and voice, which works better than any single source.

6.0 Multimodal Data Fusion for Mental Health Prediction

6.1 Rationale for Combining Multiple Modalities

Emotional and psychological state is complex, and one data source cannot capture all of it. Studies report that combining voice, text and facial analysis with machine learning improves accuracy and reliability compared with using one modality (Biomedical and Pharmacology Journal, 2025b).

6.2 Fusion Strategies

There are a few ways to combine modalities:

- **Early (feature-level) fusion:** merge raw or lightly processed features before classifying.
- **Late (decision-level) fusion:** train one model per modality, then combine their outputs.
- **Hybrid fusion:** a mix of the two.

One study fused text, audio and visual data using CNNs for emotion analysis and beat the single-modality approaches (npj Mental Health Research, 2024). Another used a memory fusion network to combine verbal, visual and acoustic streams (Scientific Reports/PMC, 2025). More recent work uses attention to adjust how much each modality counts, and graph neural networks to model how voice, text and face interact. Some report accurate results for anxiety and depression.

6.3 Existing Multimodal Mental Health Systems

Several systems show how this works in practice:

- One pipeline runs facial expressions through a CNN-LSTM, text through an LSTM and audio through another LSTM, then combines them with transfer learning and ensembling.
- Another used video, audio and text from DAIC-WOZ and E-DAIC, with VGG-16 for speech, BERT for language and ResNet-50 for vision. The fused model beat a text-only baseline, but only slightly, and text stayed the strongest single source (Scientific Reports/PMC, 2025).
- The Integrative Multimodal Depression Detection Network fused video, audio and text using a Kronecker-product mechanism. It reported that combining sources gave a fuller and more robust picture than any one alone (PMC, 2025).
- A fourth system used separate audio, video and text branches feeding one shared feature vector, designed to go beyond questionnaire-only screening (PMC, 2024c).

6.4 Machine Learning Architectures Used for Fusion

Across these systems, CNNs, LSTMs, GRUs and BiLSTMs are the usual building blocks for extracting features from each modality. Pretrained transformers like BERT (text) and Wav2Vec 2.0 (audio) are increasingly taking over that job. Attention-based or graph-based layers then handle the fusion into one prediction (arXiv, 2025a). Overall, this work supports the approach of this project, which is to combine facial, voice and questionnaire data into one wellness assessment (npj Mental Health Research, 2024).

7.0 Mobile Health (mHealth) Applications for Mental Wellness

7.1 Overview of Existing Mental Health Monitoring Apps

There are an estimated 10,000 to 20,000 mental health apps available, so the market is crowded (PLOS Digital Health, 2022). One analysis of 92 screening and diagnostic apps from the Apple App Store, Google Play and the mHealth Index and Navigation Database found that depression was by far the most targeted condition. Most apps screened for one condition only and were not integrated. A scoping review of AI-enabled apps found seventeen studies on apps using AI for conversational support, mood prediction and risk assessment. It also noted that solid evidence on outcomes for these AI features is still limited (PLOS Digital Health, 2022).

7.2 Comparative Strengths and Gaps

Compared with in-person care, apps are good on access, cost and privacy. PHQ-9-based apps have shown good sensitivity in finding users with high depressive symptoms across different populations (Biomedical and Pharmacology Journal, 2025a). However, most apps use only one type of input, usually a questionnaire or chat text. Very few combine face, voice and questionnaire signals in a single mobile session.

7.3 Design Considerations for AI-Based Mobile Health Applications

Three design points appear again and again. First, engagement matters, since many users drop out of screening apps. Second, there is a trade-off between running models on the device and in the cloud: on-device is faster and more private, while cloud can be more powerful. Third, risk results must be shown clearly without alarming the user, especially if the app has to warn someone about raised suicide risk.

8.0 Ethical, Privacy and Safety Considerations

8.1 Data Privacy and Security

Face video, voice recordings and questionnaire answers are all very sensitive. Because of this, the literature treats privacy, secure storage and informed consent as must-haves for any AI mental health tool. A systematic review of speech-based suicide risk assessment was direct about it: strong anonymization and clear patient consent come first, and systems also need protection against threats like data breaches before wider use.

8.2 Ethical Concerns in AI-Based Screening

A common worry is that people may treat AI output as a diagnosis when it is only a screening signal (The Hastings Center, 2024). Some researchers deal with this by presenting their models as screening assistants for human review, not diagnostic tools, especially for high-risk content such as suicide-related material. Other concerns are algorithmic bias, the difficulty of explaining black-box models in clinical settings, and the risk that normal stress gets read as a disorder when social context is ignored (Taylor & Francis, 2026).

8.3 Responsible Handling of Suicidal Ideation Risk

Ethical guidance for AI suicide risk tools mostly agrees on a few principles:

- collect only the minimum data needed;
- explain clearly how data will be used and stored;
- use strong encryption and access controls;
- keep a human involved in any decision to escalate;
- leave the final clinical judgement to qualified professionals.

Analyses also stress informed consent, respect for autonomy and dignity, and openness about the fact that risk screening is happening, since users may not expect their data to be checked for anything beyond why they opened the app (The Hastings Center, 2024; Taylor & Francis, 2026).

9.0 Research Gap and Summary

9.1 Synthesis of Findings

Three main points stand out. First, facial and voice analysis have each been shown to be workable and fairly objective ways to learn about depression and anxiety, supported by deep learning research and clinical datasets like DAIC-WOZ. Second, questionnaires such as the PHQ-9 remain reliable, and moving them to a phone does not seem to reduce their accuracy. Third, combining face, voice and text or questionnaire data regularly beats any single source, and this is the technical basis of this project.

9.2 Identified Gaps

- Most multimodal research uses offline datasets like DAIC-WOZ, not real-time, mobile-first apps.
- Most commercial apps still use a single input, usually a questionnaire or a text chatbot.
- Face and voice datasets are small and not very diverse, for good privacy reasons. This limits how well models generalise and makes careful data handling a requirement from the start (ScienceDirect, 2025a).
- Ethical guidance for suicide risk screening is well developed in principle, but few real systems show how ideas like minimal data collection and clear consent work end to end inside one consumer app.

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